BRIEF DESCRIPTION OF THE PLASMA SCENARIO AND EXPERIMENTAL DATA

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TORPEX has major radius R = 1 m and minor radius a= 0.2 m. In the experiments investigated here, hydrogen plasmas are produced and sustained by microwaves in the electron cyclotron range of frequencies. A microwave power of ~400 W is used and the plasma source is localized on the high field side, with negligible plasma production for r > 5 cm. A toroidal magnetic field of ~76 mT on axis and a vertical magnetic field B_z=2.3 mT are used, which results in a simple magnetized torus (SMT), characterized by vertically elongated plasma profiles.

This SMT configuration has been extensively studied in a number of papers:

M. Podestà et al., Plasma Phys. Controlled Fusion 47, 1989 (2005).
I. Furno et al., Phys. Plasmas 15, 055903 (2008).
I. Furno et al., Phys. Rev. Lett. 100, 055004 (2008)
C. Theiler et al., Phys. Rev. Lett. 103, 065001 (2009).

Time averaged profiles

The time averaged profiles of the plasma density n_e, the electron temperature T_e, and the plasma potential V_{pl} are obtained from Langmuir probe measurements. The first two quantities and the floating potential V_{fl} are extracted from fits of Langmuir probe characteristics. The plasma potential is deduced from the general formula: $V_{pl}=V_{fl}+\Lambda T$, where Λ=3.1 is evaluated experimentally. The time averaged 2D profile of the electron density and electron temperature are shown in Fig. 1. Profiles are peaked around r=−2 cm and elongated in the vertical direction.

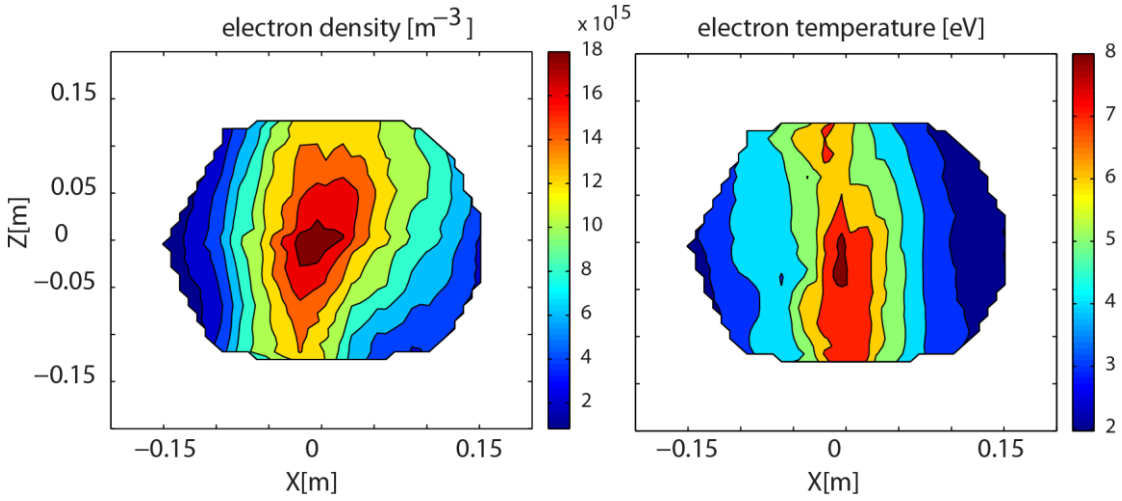


Figure 1: experimental profiles of electron density and temperature

This configuration is characterized by two distinct poloidal regions with different plasma dynamics: 1) a main plasma region for −3<r<6 cm dominated by a coherent interchange wave; 2) a region on the low-field side for r > 6 cm with negligible plasma production, dubbed source free region, characterized by the propagation of plasma blobs, resulting in intermittent transport events.

The profiles of Fig. 1 are obtained by scanning an array of LPs with spatial resolution of approximately 1 cm in radial and vertical direction. Due to a limited diagnostics access, the measurements are limited to approximately X = [-0.15, 0.15] m and Z = [-0.13, 0.13] cm. In order to provide the electron density over the entire poloidal cross section of the device we have interpolated the electron density at the midplane (Z=0), $N_{e,mid}$. This is reasonably well interpolated by the following function:

$$N_{e,mid} = a2*\exp(-((X-b2)/c2).^2) + a3*\exp(-((X-b3)/c3).^2),$$

where:

$$\begin{aligned} a2 &= 9.063e+15 \\ b2 &= -0.019 \\ c2 &= 0.060 \\ a3 &= 1.13e+16 \\ b3 &= 0.030 \\ c3 &= 0.124 \end{aligned}$$

The electron density is then linearly interpolated over the entire poloidal cross section of TORPEX assuming a value of the density at the edge of the device at radius =0.2 m, $n_{e,edge}(X_0,0) = N_{e,mid}(X_0,Z_0)/2$. Note that the 0.5 factor boils down to the assumption of a classical pre-sheath density drop corresponding to Bohm boundary conditions at the edge of the device. The electron density profile is shown in Fig. 2 below:

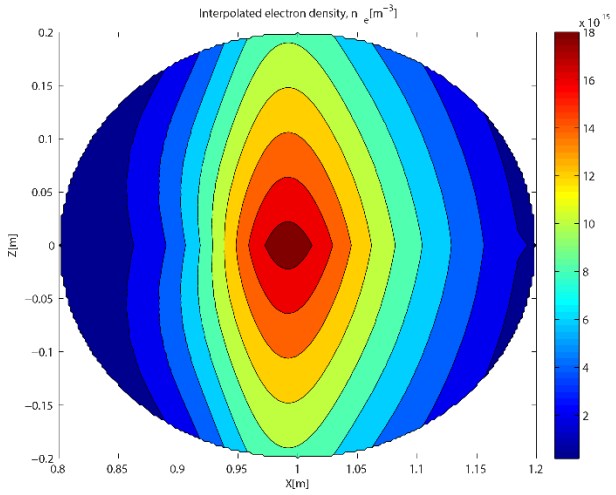


Figure 2: interpolated electron density profile

The data corresponding to this profile are contained in the file `ne.txt`. The format for the data is the same as for the magnetic field as explained below. The spatial resolution is 1 mm in both directions.

Magnetic fields of the SMT

Following a brief discussion with L. Figini – Milan (see also attached his notes on the reference system used in GRAY), we are using the following format for the magnetic fields:

```
n  m
R1  R2  ...  Rn
z1  z2  ...  zm
X(R1, z1)  X(R2, z1)  ...  X(Rn, z1)
X(R1, z2)  X(R2, z2)  ...  X(Rn, z2)
...
X(R1, zm)  X(R2, zm)  ...  X(Rn, zm)
```

where X are B_R, B_ϕ, B_z, n_e.

The data are saved in the following ASCII files:

```
B_Tbc_y.txt (toroidal component)
B_V_z.txt (vertical component)
B_V_x.txt (radial component)
```

The different components are shown in Fig. 3 below. The spatial resolution is 1 mm in both directions.

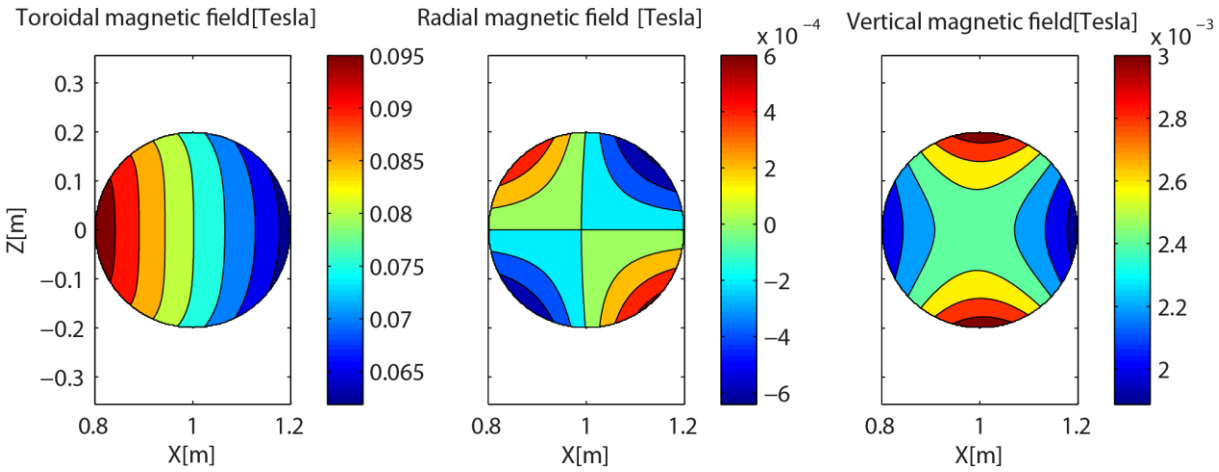


Figure 3: the three components of the magnetic field

For those of you running Matlab the different plots of this document can be produced by running the routine: `plot_profiles.m`